Workshop Pole Barn: Sections and Construction Details

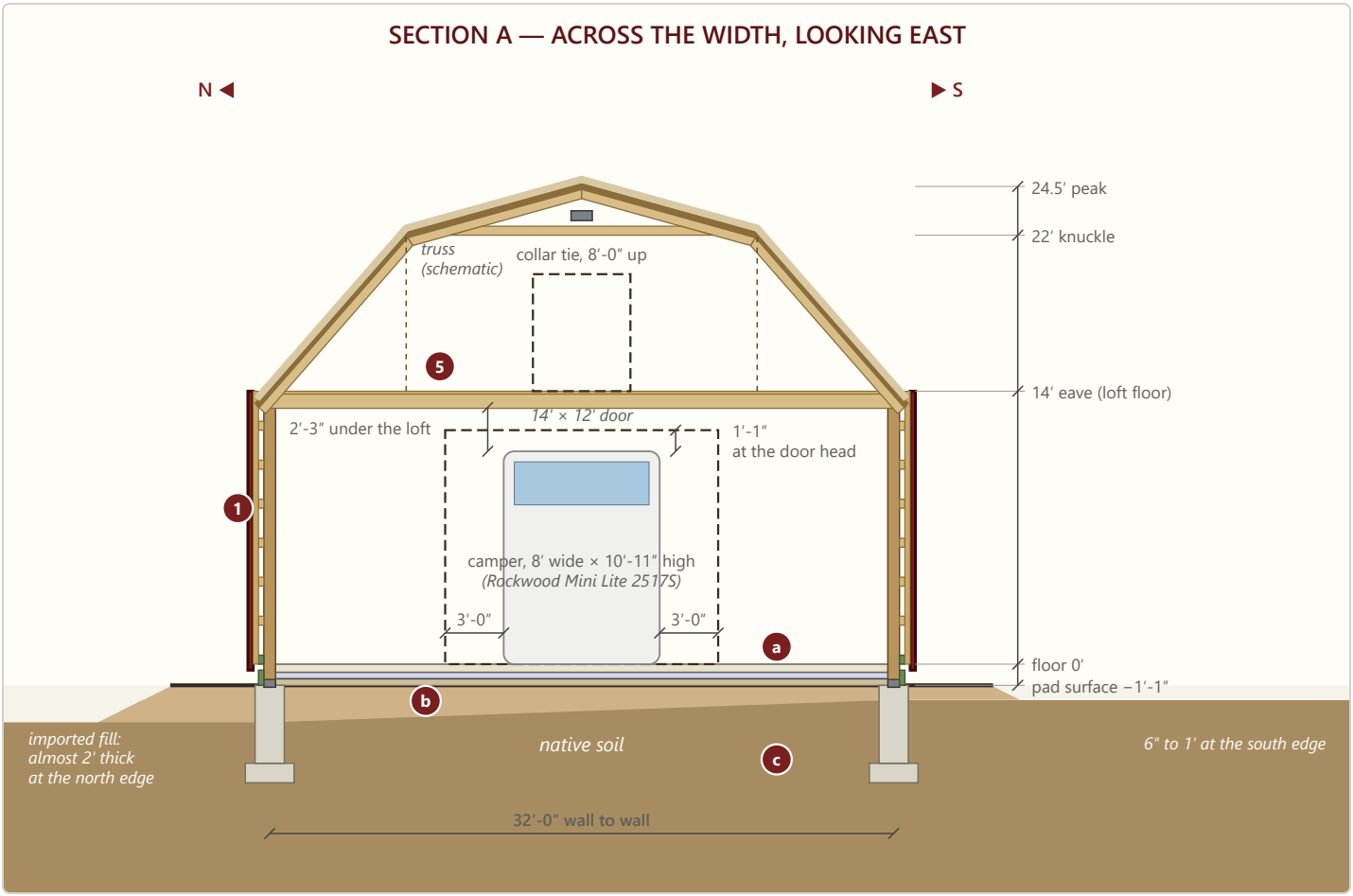
Cross-sections through the building, the two framing plans, and seven layer-by-layer details · 32 × 48 ft · 14 ft eave · gambrel

These are design-intent drawings for the engineer, the truss plant, and the builder. Truss members and connections are placeholders until the truss designer and the engineer size them. Layers in the details are exaggerated where they say so. The markers match the tables under each drawing.

What Is in This Section

1. **Section A, across the width**, looking east: the gambrel truss with its collar tie, the loft floor, the camper in the door, and the pad's fill under the slab.
2. **Section B, along the length**, looking north: the loft floor, the collar ties, the stairs, the main door, the hoist beam and one of its braces, and the deck.
3. **Plan F1, foundation and base framing**: piers, posts, the splash board, and the slab joints, on the pad.
4. **Plan F2, loft floor framing**: the 25 truss bottom chords at 2′ OC, the stair opening, the loft door header, the hoist beam and its braces, and the deck.
5. **Detail 1** wall assembly · **2** roof assembly · **3** pier and post base · **4** rainscreen base · **5** eave vent-out · **6** loft door, hoist beam, and deck · **7** truss bearing.
6. **Detail 5 in perspective**: the wall and the eave cut through a post and its truss, with the rainscreen's air path from the base to the eave and the heel block that joins the two chords.

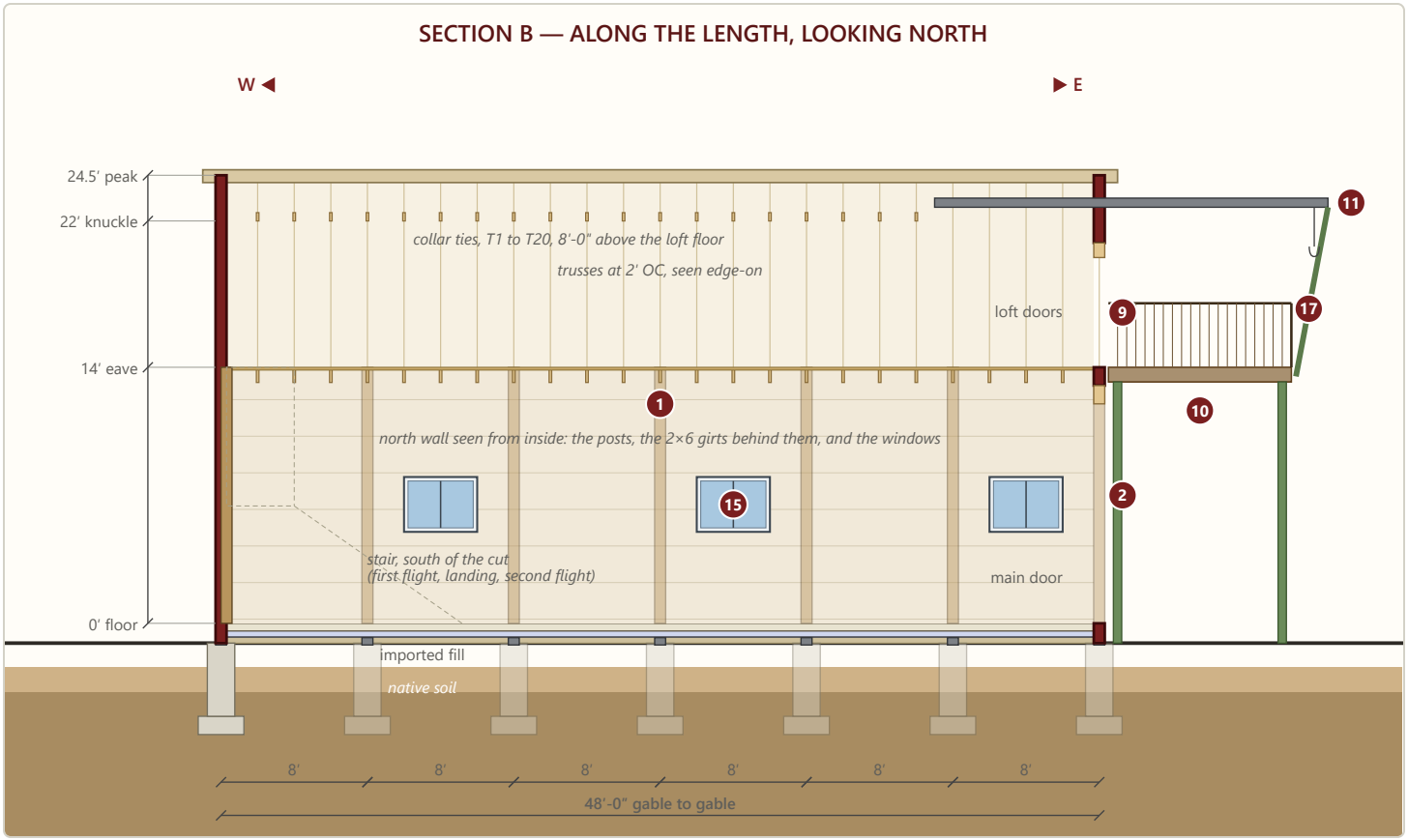
Section A: Across the Width (looking east)



Section A cuts across the 32 ft width and looks east at the gable wall. The 14′ × 12′ door and the loft doors on the east gable are drawn dashed behind the cut. The camper (10′-11″ tall as listed) has 1′-1″ under the door head and 2′-3″ under the loft floor. The truss is a schematic envelope: its chords and the collar tie across the upper chords, 8′-0″ above the loft floor, without the webs, the plates, or the heel block (the truss package draws those).

Marker	What it shows
1	3-ply 2×8 laminated post on its pier, with the girts and the wall layers outside it (sheathing, rainscreen, steel) and the splash board at the base
5	¾″ T&G plywood loft floor on the truss bottom chords, 14′ above the floor
a	The floor stack: 5″ slab on 4″ rigid foam and a 10 mil vapor barrier on 4″ compacted gravel, about 1′-1″ above the pad surface
b	Imported fill, the level pad: almost 2′ thick at the north edge and 6″ to 1′ at the south edge. The shoulders are drawn at 2:1 and are not surveyed
c	Native soil. The pier pads bear here, about 3′ down into it at the north row and 4′ to 4½′ at the south row

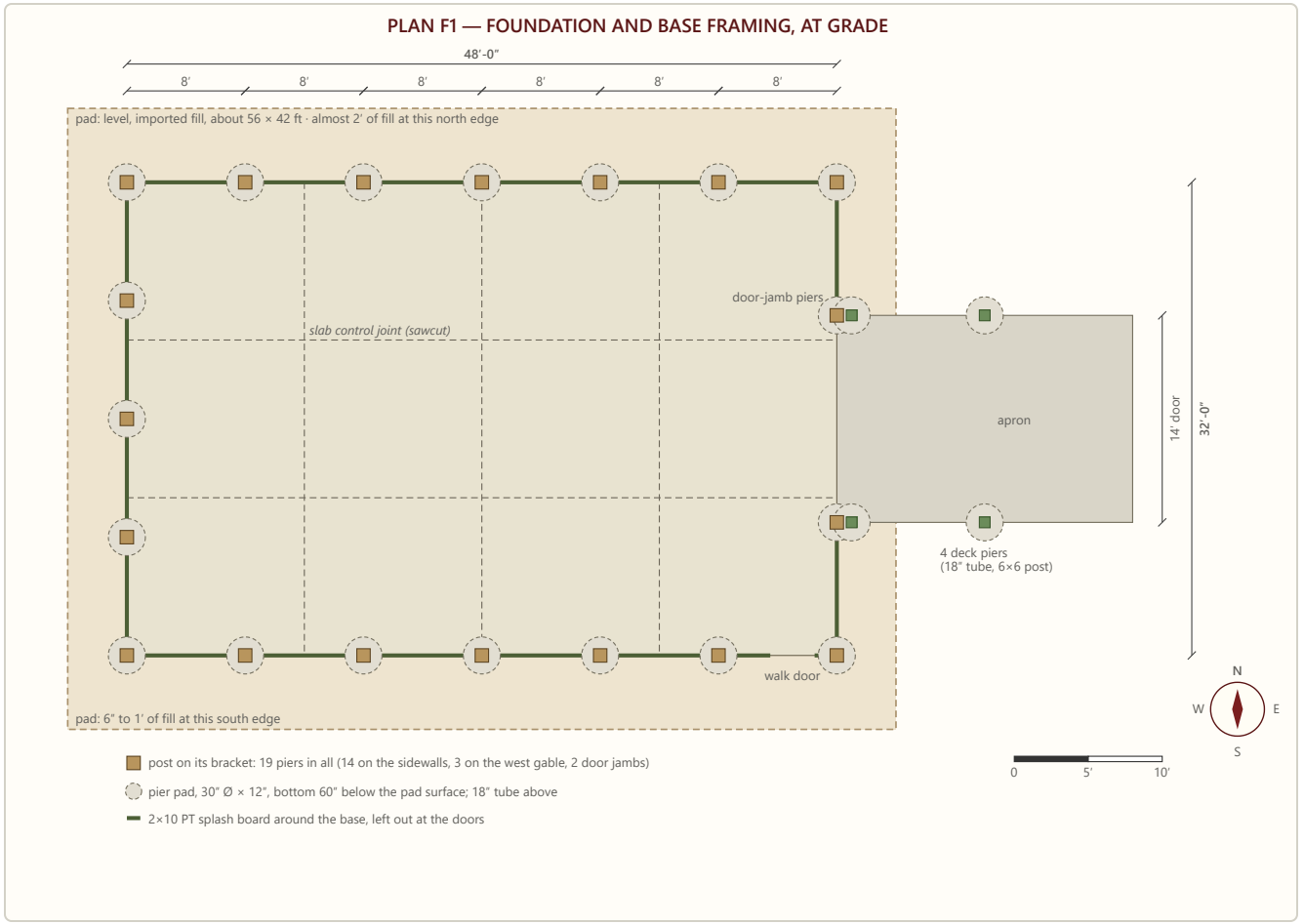
Section B: Along the Length (looking north)



Section B cuts along the 48 ft length at the center line and looks north. Sidewall posts are at 8-8-8-8-8-8 ft; the trusses (2′ OC) are seen edge-on, and the collar ties on T1 to T20 are cut at the center line; the loft floor and the ridge are cut. The east gable shows the main door, the loft doors, the hoist beam, one of its two braces, and the deck. The stair is south of the cut, so it is drawn dashed.

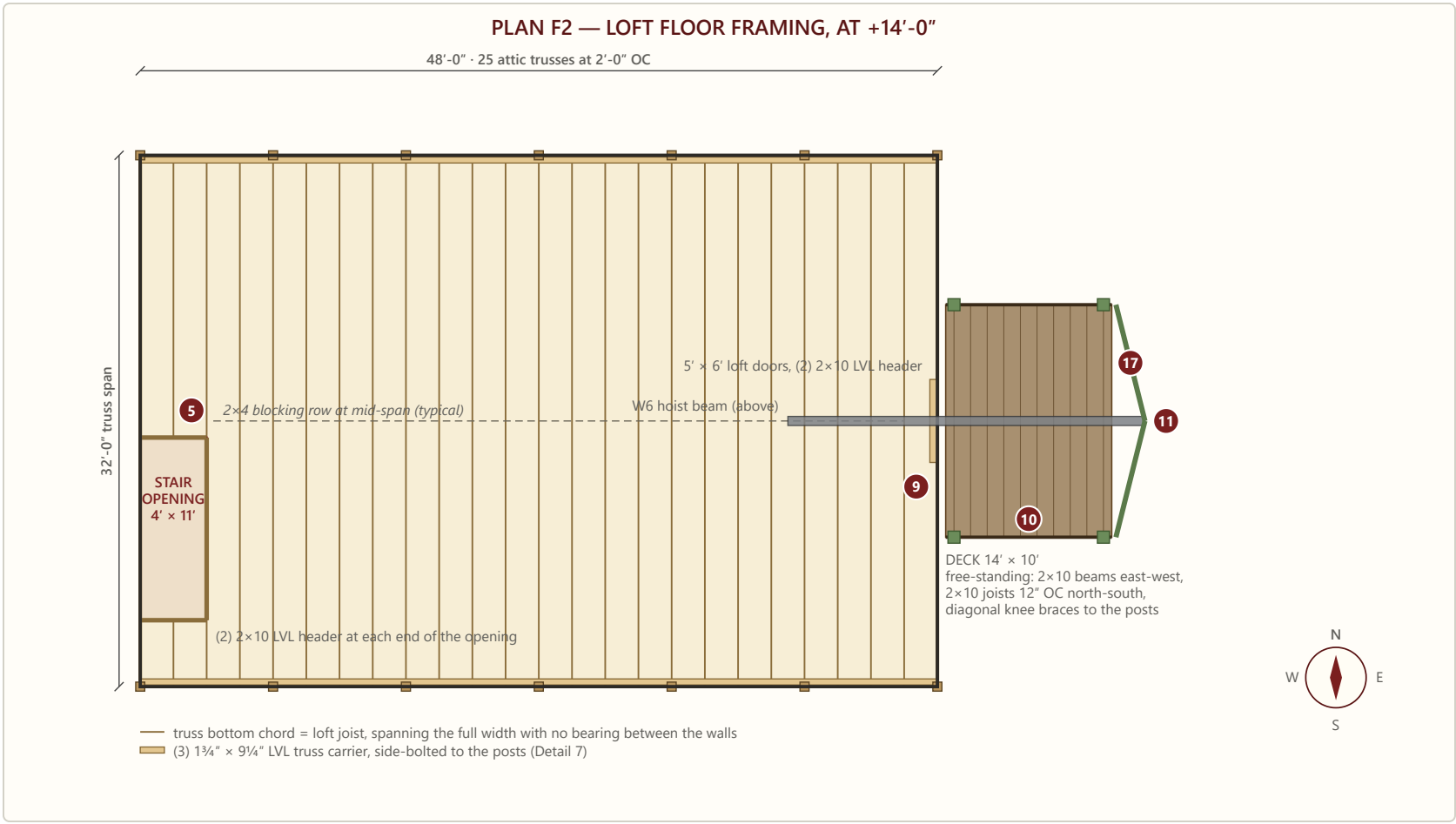
Marker	What it shows
1	Sidewall post (the north wall seen from inside, with its 2×6 girts behind the posts)
2	The 14′ × 12′ main door opening in the east gable, with its LVL header above
9	The 5′ × 6′ loft doors at the loft floor, with their header
10	The 14′ × 10′ landing deck, 10′ deep, on posts to grade
11	The W6 hoist beam through the east wall: 12′-6″ out, 9 ft in, with the chain fall near its tip
17	A hoist brace, from the deck's outer corner to the beam's tip (the other is on the near side of the cut)
15	The north wall's windows, 4′ × 3′, sill 5′ above the floor, centered between the posts in bays 2, 4, and 6

Plan F1: Foundation and Base Framing



Plan F1, at grade. **19 building piers** (14 on the sidewalls at 8-8-8-8-8-8, 3 on the west gable, 2 door-jamb piers) plus 4 deck piers. All are 30″ Ø × 12″ pads, bottom 60″ below the pad surface.

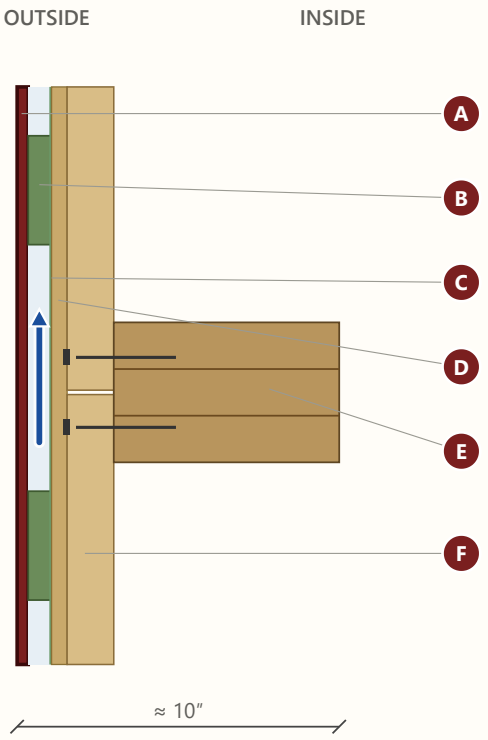
Plan F2: Loft Floor Framing



Plan F2, at +14′-0″. **25 attic-truss bottom chords at 2′ OC act as the loft joists**, spanning the full 32 ft width with no bearing between the walls. Floor build-up: ¾″ T&G plywood, glued and screwed, its long edge across the chords; foam or insulation between the chords is optional because the loft is conditioned on both sides.

Detail 1: Wall Assembly

DETAIL 1 — WALL ASSEMBLY (horizontal section, layers exaggerated)



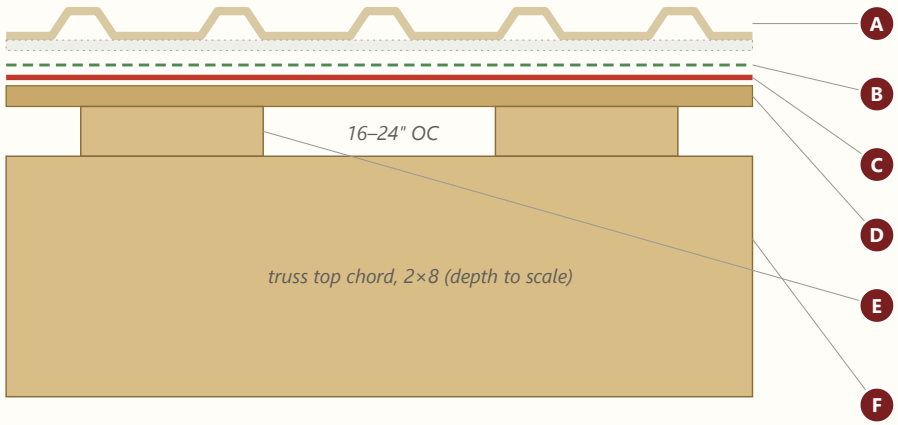
The furring stands off the sheathing to make a ¾″ air space that drains and dries (blue arrow: air moving up). The girt is a 2×6 laid flat on the post. Its joints fall on post center lines, and screws go through it into the post.

Marker	Layer
A	26 ga steel siding, vertical panels, barn red, fastened to the furring with #10 × 1½″ hex screws with EPDM washers
B	1×4 PT furring at 24″ OC, vertical: makes a ¾″ ventilated rainscreen cavity
C	Tyvek CommercialWrap, taped seams, shingle-lapped, fastened with cap nails
D	½″ OSB sheathing over the girts, APA Exposure 1, 8d ring-shank nails at 6″ on the edges and 12″ in the field (4″ at corners and openings)
E	3-ply 2×8 laminated PT post at 8′ OC, on a wet-set bracket (Detail 3)
F	2×6 SPF #2 girt, laid flat on the post (1½″ thick, 5½″ tall), 24″ OC, continuous over the posts with its joints on post center lines, as the two boards meeting here do. Each board is screwed to the post (the dark pins): 2 structural screws at a board end, 3 where a board crosses a post

Detail 1, a horizontal slice through the wall at a girt joint over a post, layers exaggerated. The girt is laid flat on the post, and the sheathing goes over it. The ¾″ furring cavity is the most important detail in the wall: it lets the wall dry behind the steel and keeps moisture from reaching the sheathing.

Detail 2: Roof Assembly

DETAIL 2 — ROOF ASSEMBLY (section at a purlin, lower slope, shown flat)

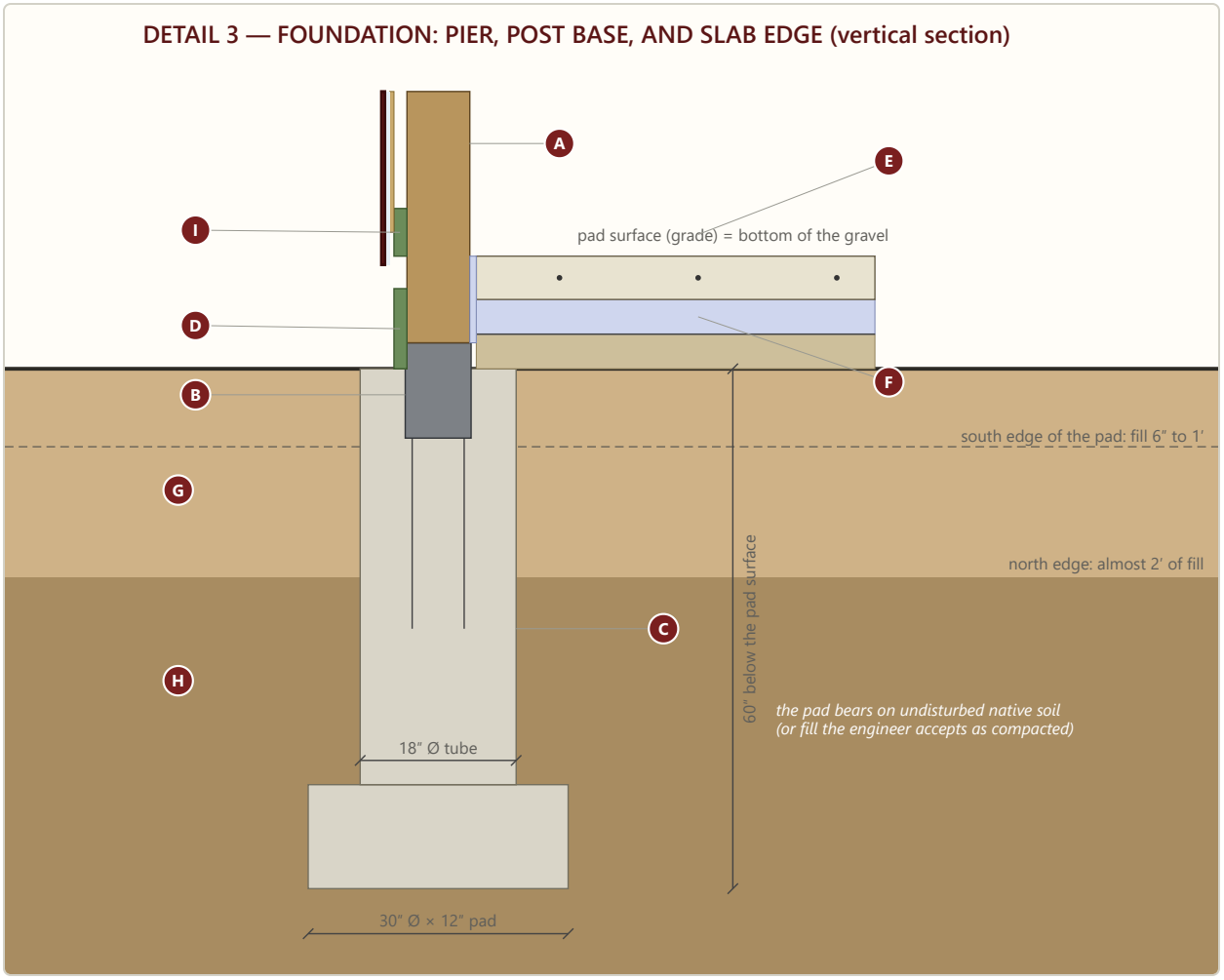


Layers are exaggerated. The lower slope is about 49° and gets full ice and water shield; the upper slope (16°) gets it only at the pitch break and around penetrations. Dripstop on the steel's underside controls condensation in the loft.

Marker	Layer
A	26 ga steel roof panel, factory Dripstop on the underside, beige, cut to the full slope length; #10 × 1½" hex screws with EPDM washers in the manufacturer's high-snow pattern
B	Synthetic underlayment (Titanium UDL30 or equal) over the entire roof
C	Ice and water shield: FULL on the lower (steep) slope; at the pitch break and around penetrations on the upper slope
D	⅝" OSB roof deck with H-clips between the purlins
E	2×6 SPF purlins on top of the truss chords, 16–24" OC per the truss design
F	Pre-engineered gambrel attic truss, 32' span, 2' OC, PE stamped for 120 psf ground snow

Detail 2, the stack-up on the lower (steep) gambrel slope, shown flat with the layers exaggerated. The real stack is under 4" thick, plus the truss chord. The snow guards are mounted separately over the walk door.

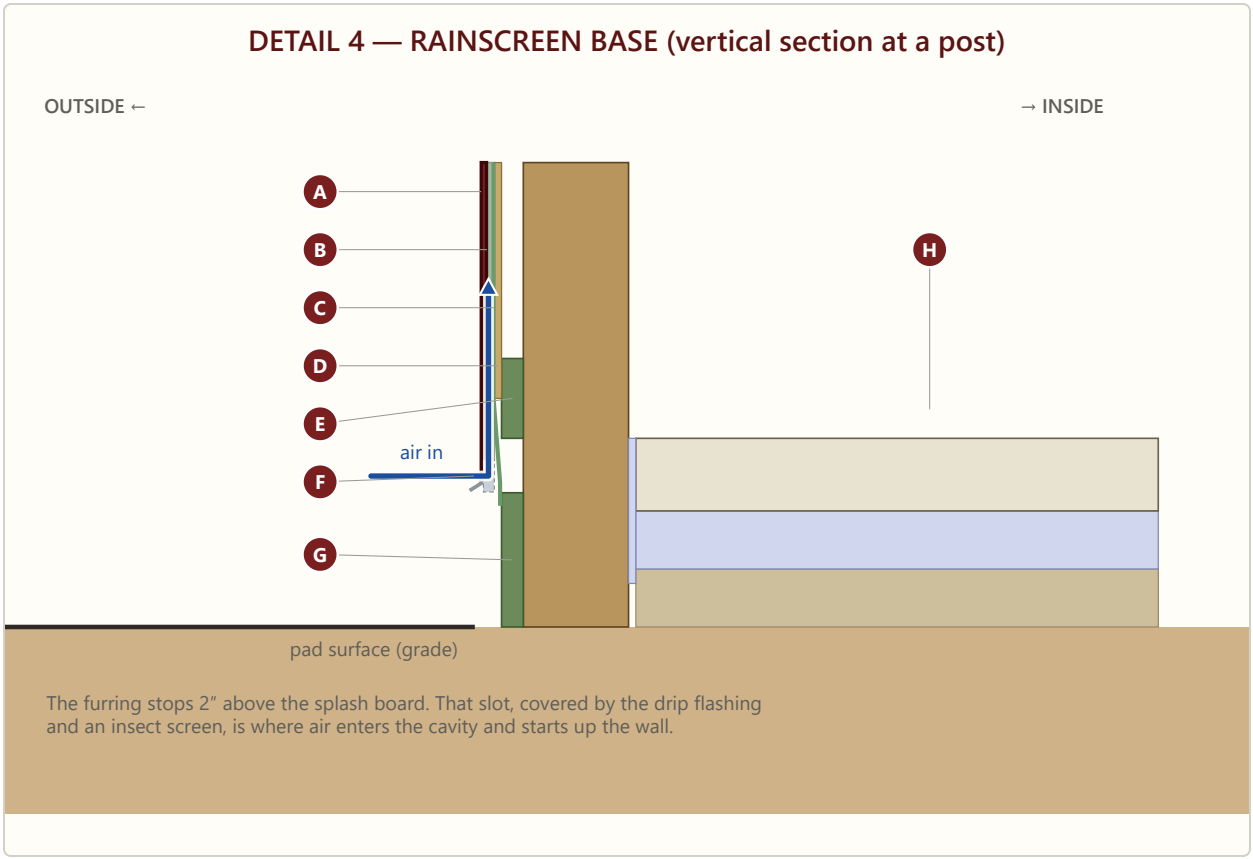
Detail 3: Foundation, Pier, and Post Base



Detail 3, a vertical section through a sidewall post, drawn at the north edge of the pad where the fill is thickest. The wet-set bracket lifts the post above the concrete, so wood never touches the ground. The pier passes through the imported fill and bears on native soil. The slab stands on its own, with an isolation strip against the post.

Marker	What it is
A	3-ply 2×8 laminated post (4½" × 7¼")
B	Perma-Column or Sturdi-Wall wet-set bracket, set in the wet pier concrete
C	4,000 psi concrete pier: an 18" tube on a 30" Ø × 12" pad, bottom 60" below the pad surface (how to dig and form it) (the county's 30" below the frost line)
D	2×10 PT splash board, screwed to the posts in the same layer as the girts
E	5" slab, #4 rebar at 16" OC each way
F	4" rigid foam, 10 mil vapor barrier, and 4" compacted gravel, on the compacted fill
G	Imported fill, the level pad: up to about 2' at the north edge, 6" to 1' at the south edge
H	Native soil: the pier pad bears here, or on fill the engineer accepts
I	2×6 PT base girt, laid flat on the post at the floor line (13" above the pad): the lowest girt row, which backs the lower edge of the sheathing

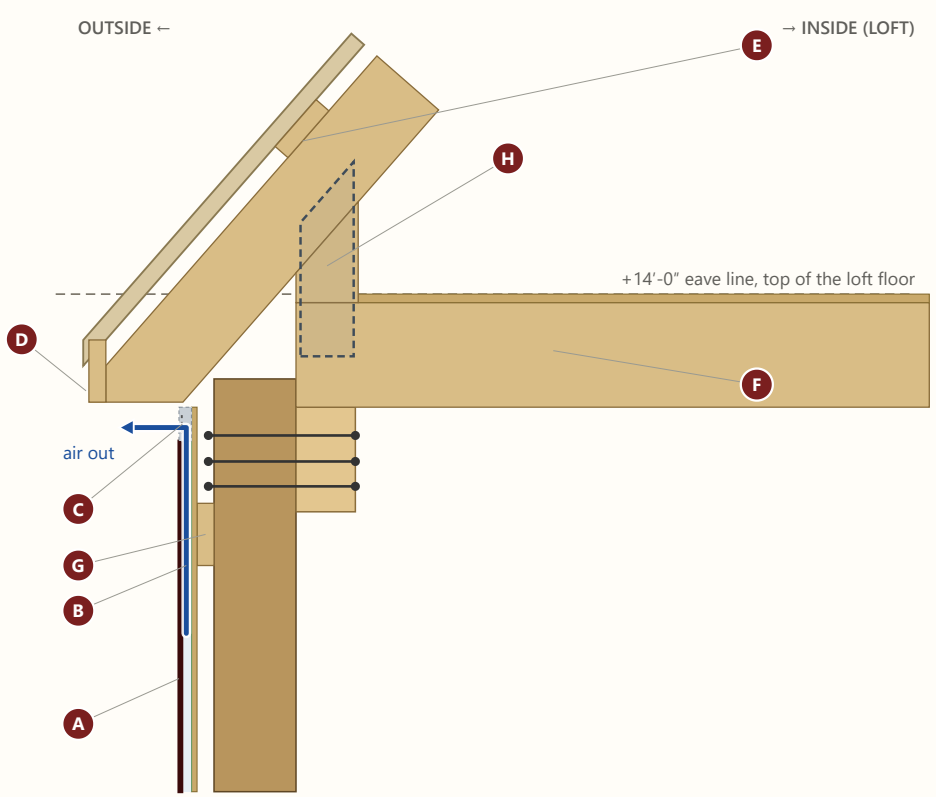
Detail 4: Rainscreen Base



Detail 4, the base of the wall where the rainscreen takes in air. The wrap shingle-laps over the top of the splash board. The furring stops short of the splash board, leaving a vent slot covered by the drip flashing and an insect screen. Air enters here, rises through the cavity, and leaves at the eave (Detail 5).

Marker	What it is
A	26 ga steel panel, bottom edge about 1" above the splash board's top
B	1×4 PT furring, stopping 2" above the splash board
C	Tyvek WRB, lapping over the top of the splash board, shingle style
D	½" OSB sheathing over the girts, its lower edge on the center line of the base girt
E	2×6 PT base girt at the floor line, laid flat on the post: the lowest girt row, which backs the sheathing's lower edge
F	Drip flashing and insect screen: the rainscreen's air inlet
G	2×10 PT splash board, screwed to the posts in the girt layer, keeping the siding and sheathing off the ground and the snow
H	5" concrete slab on foam and gravel

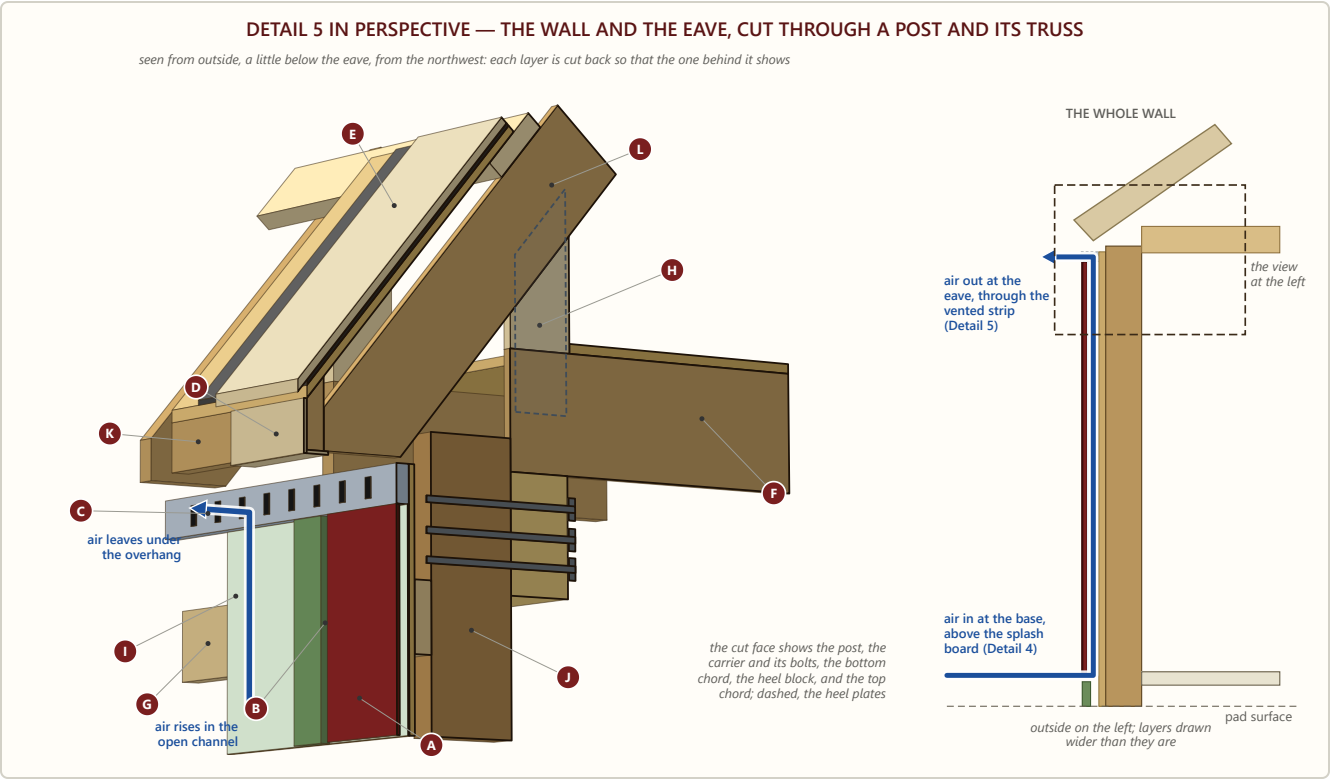
Detail 5: Eave (the Rainscreen Vents Out)



Marker	What it is
A	26 ga steel siding, stopping under the vented closure strip
B	1×4 PT furring, vertical and continuous up to the closure strip. The section is cut between two strips, so the cavity shows as an open channel
C	Vented closure strip: a 3″ perforated strip with an insect screen across the top of the cavity, in line with the steel and under the overhang. It is the rainscreen's air outlet
D	Eave trim: wraps the sub-fascia and laps under the roof steel's edge
E	Roof assembly on the truss top chord: steel, underlayment, ice and water shield, OSB, and purlin
F	Truss bottom chord and ¾″ T&G loft floor
G	2×6 girt, the highest row of the wall (its bottom edge 12′-0″ above the floor), laid flat on the post under the sheathing
H	Heel block: a 2×6 on end between the top chord and the bottom chord at the bearing, with the truss plant's steel plates over it on both faces (dashed: behind the cut). The loft floor butts to the block

Detail 5, a vertical section where the rainscreen vents out at the eave. Air that entered at the splash board (Detail 4) rises through the furring cavity, passes through the vented closure strip across the top of the cavity, and leaves under the overhang. Without that top vent the cavity becomes a dead-air pocket that traps moisture. The heel block (H) joins the top chord to the bottom chord. The next section shows the same joint in perspective.

Detail 5 in Perspective: How the Wall and the Eave Tie Together

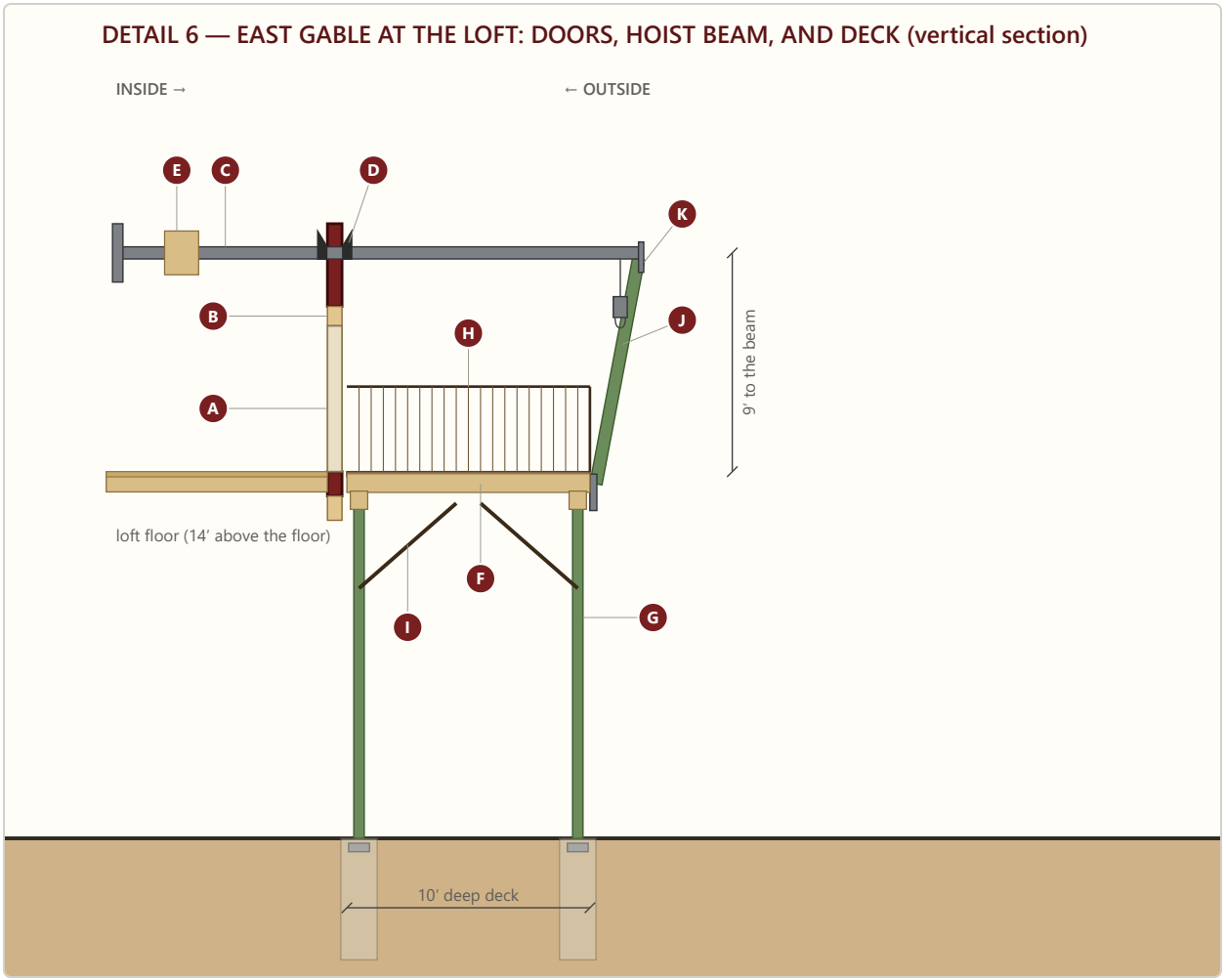


The north wall at the eave, cut through a post and the truss over it, seen from outside and a little below the eave. Each layer is cut back so that the one behind it shows. The blue path is the rainscreen's air: it enters at the base above the splash board (Detail 4), rises through the open channels beside the furring, passes through the vented closure strip, and leaves under the overhang. The dashed outline is the steel plates on both faces of the heel, behind the cut, and the diagram at the right shows the whole wall. The letters A to H match Detail 5, and I to L show only here.

Marker	What it shows
A	26 ga steel siding, stopping under the vented closure strip. It is cut back here to show the furring
B	1×4 PT furring, vertical at 24″ OC (one strip shows). It holds the steel off the wall, and the air rises beside it
C	Vented closure strip: 3″ tall, perforated, with an insect screen, in line with the steel. The air leaves through its slots under the overhang
D	Eave trim: bent metal that wraps the sub-fascia and laps under the roof steel's edge
E	Roof on the truss top chord: 2×6 purlin, ⅝″ OSB deck, ice and water shield with underlayment, and 26 ga steel, each cut back farther than the one above it
F	Truss bottom chord (the loft joist) with the ¾″ T&G loft floor on it, which butts to the heel block

Marker	What it shows
G	2×6 top girt, laid flat on the post's outside face under the sheathing
H	Heel block: a 2×6 on end between the top chord (L) and the bottom chord (F) at the bearing. The truss plant's steel plates are pressed over it, and onto both chords, on both faces (dashed)
I	½″ OSB sheathing with the housewrap over it: the wall behind the cavity, and the air barrier
J	3-ply 2×8 post, cut through, with the LVL truss carrier on its inside face and (3) ½″ through-bolts. The bottom chord lands on the carrier
K	2×6 sub-fascia closing the plumb-cut end of the truss tail
L	2×8 truss top chord, cut through. Its tail has a plumb cut for the sub-fascia and a level cut under it, so the air leaves into the open. At the bearing it rests on the heel block

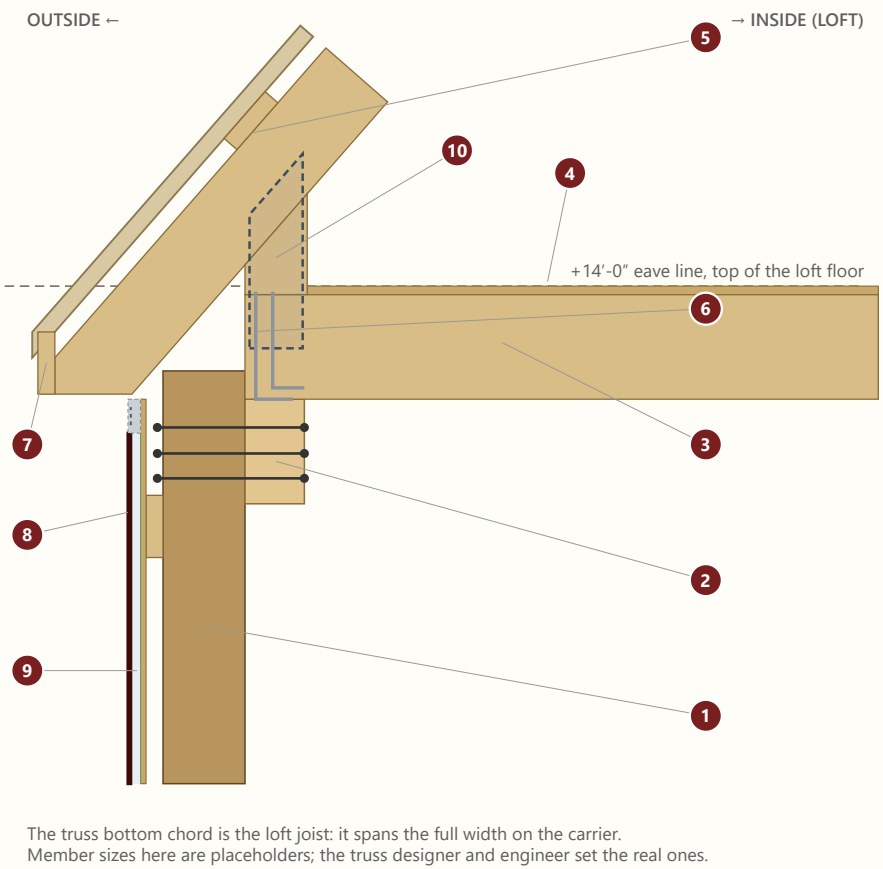
Detail 6: Loft Door, Hoist Beam, and Landing Deck



Detail 6, a vertical section through the east gable at the loft. The loft doors have an LVL header; the W6 hoist beam passes through the wall and out past the deck's outer edge, anchored to the truss top chord inside with a weather boot at the penetration. A 6×6 brace rises from a steel shoe on the end of each outer deck beam to a plate on the beam's tip (one brace is shown, beyond the cut). The free-standing 14′ × 10′ deck stands on 6×6 posts to pier footings, braced with diagonal knees.

Marker	What it is
A	5′ × 6′ insulated double doors at the loft floor, locking from inside, opening clear for the hoist
B	(2) 2×10 LVL header above the loft doors (engineer-sized)
C	W6×12 steel hoist beam, 21′-6″: 9 ft inside and 12′-6″ out, 2 ft past the deck's edge and 9′ above the loft floor, rated for 500–1,000 lb. The engineer picks the size
D	EPDM weather boot and flashing where the beam passes through the wall: a CRITICAL detail
E	Anchor plate and bolt-through hardware to the truss top chord, with blocking between trusses to spread the load. The truss manufacturer reviews it
F	Deck framing: doubled 2×10 PT beams, 2×10 PT joists at 12″ OC, decking on top
G	6×6 PT posts on 18″ Ø piers, 60″ below grade, in post bases
H	42″ railing (fall protection above 30″)
I	Diagonal knee braces: a 14 ft tall deck will sway without them
J	Hoist brace: 6×6 PT, about 11′-9″, from a steel shoe on the end of the outer deck beam up to the beam's tip. A second brace mirrors it on the other side. The rail is framed around them
K	Steel plate on the beam's tip, with an ear for each brace; it is also the trolley's outer stop. The hook hangs a few inches in from it

Detail 7: Truss Bearing at the Top of a Post



Marker	What it is
1	3-ply 2×8 laminated post (4½″ × 7¼″) on a wet-set bracket below
2	(3) 1¾″ × 9¼″ LVL truss carrier, side-bolted to the post with (3) ½″ galvanized through-bolts per post
3	Truss bottom chord (2×10, a placeholder size), bearing on the carrier and acting as the loft joist across the span
4	¾″ T&G plywood loft floor, long edge across the chords
5	Truss top chord (2×8, a placeholder size), the lower gambrel slope rising to the knuckle
6	H2.5A hurricane ties, 2 per chord bearing, one on each side
7	2×6 sub-fascia closing the plumb-cut end of the truss tail (about 12″ of overhang), with the eave trim wrapped around it
8	Wall steel stops under the vented closure strip that crosses the top of the rainscreen cavity
9	Wall assembly outside the post: 2×6 girt, OSB, wrap, rainscreen, and steel (Detail 1)
10	Heel block (a 2×6 on end) between the top chord and the bottom chord at the bearing, with the plant's steel plates on both faces (dashed: behind the cut). The loft floor butts to the block. The heel stands 13″ above the floor at the bearing's outer edge, and at least 12″

Detail 7, a transverse section through one sidewall post at the eave. The gambrel attic truss's bottom chord **is** the loft joist: it spans the full 32 ft with no bearing between the walls and lands on a **(3) 1¾″ × 9¼″ LVL truss carrier** side-bolted to the inside face of every post. A heel block joins the top chord to the bottom chord at the bearing. The member sizes are placeholders.

Mountain Prairie Acres · Workshop pole barn sections and details · updated October 23, 2026 · drawn from `docs/workshop_model.js`.